

2 (10 points)

2) for the following single flow graph find

1) Forward paths 2/2.5

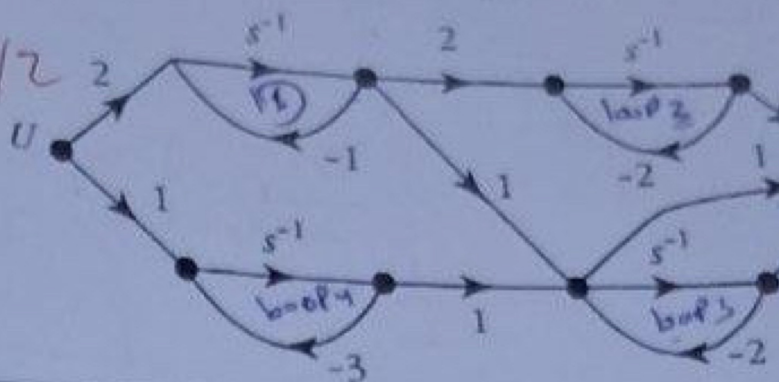
2) Feedback loops (close loops) 2/2

3) non touch loops 1.5/5

4) find Δ

5) find $y(s)/x(s)$

10 marks



1) Forward Path ($2 \times s^{-1} \times 2 \times s^{-1} \times -1$)

2) forward Path 2 ($2 \times s^{-1} \times 1 \times 1$)

3) forward Path 3 ($1 \times s^{-1} \times 1 \times s^{-1} \times 1$)

4) forward Path 4 ($1 \times s^{-1} \times 1 \times 1$)

X →

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الـ (4.5)

6.5
10

Feedback loop

loop 1 ($-1 \times s^{-1}$)

loop 2 ($s^{-1} \times -2$)

loop 3 ($s^{-1} \times -2$)

loop 4 ($s^{-1} \times -3$)

loop 5

non touch loop (two loops)

Loop 1 with loop 2

loop 1 with loop 3

loop 1 with loop 4

non touch loop

Loop 1 with 2

loop 1 with 1

~~loop 3 with 1~~

loop 3 with 1

$$\Delta = 1 - (-1 \times s^{-1} + s^{-1} \times -2 + s^{-1} \times -2 + s^{-1} \times -3) +$$

$$+ (-1 \times s^{-1} \times s^{-1} \times -3) - (-1 \times s^{-1} \times s^{-1} \times -2 \times s^{-1} \times -2)$$

non touch

+

Path 1

Path 2

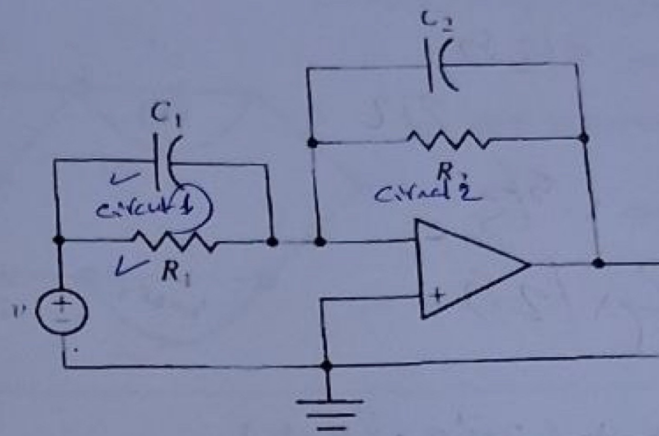
$$\Delta 1 = 1$$

$$\Delta 2 = 1$$

$$\text{transfer function} = 2 \times s^{-1} \times 2 \times s^{-1} \times -1 + 2 \times s^{-1} \times 1 \times 1 +$$



for the electrical system shown below find the transfer function of the system $V_o(s)/v(s)$



cut ① C_1 with R_1 parallel

$$\frac{1}{Z} = \frac{1}{\frac{1}{Cs}} + \frac{1}{R} = \frac{Cs}{R} + \frac{1}{R} = \frac{RCS+1}{R}$$

R_1 [1] قوة مقادير

cut ② $\rightarrow C_2 //$ parallel R_2

$$= \frac{1}{\frac{1}{Cs}} + \frac{1}{R} = \frac{Rcs}{R} + \frac{1}{R} = \frac{R}{RCS+1} = Z$$

transfer function = $-\frac{V_o}{V} = -\frac{R}{RCS+1} = \frac{R}{RCS+1}$

the answer equal -